SMART INDIA HACKATHON 2026 · PS 26158 · NTRO

Single-Pass Drone Video to Accurate 3D Models

Research report — who has solved what, how, with what accuracy, and what remains open. Key finding: *GPS anchoring turns single-pass video metric*.

Project	Drishti3D
Team	Taraket
Problem statement	26158 · Single-Pass Drone Video to Accurate 3D Model Generation System
Date	30 September 2026

CONTENTS

1. No product ships single-pass video to metric 3D today
2. Georeferencing, not reconstruction, sets the accuracy ceiling
3. GPS-anchored feed-forward models beat pure SLAM on single passes
4. Gaussian splats look better but measure worse
5. Open-source parts exist but are batch-grade or abandoned
6. India's constraints favour an offline, CORS-aware tool
7. A buildable architecture that wins on accuracy engineering
8. Conclusion

Every number carries its source link. Full notes, bibliography and design docs: https://github.com/vikranthsai310/sih2026-2

GPS anchoring turns single-pass video metric

Nobody has fully solved PS 26158. No product or paper, as of 30 September 2026, takes one monocular drone video pass with consumer GPS and returns a validated, georeferenced, textured mesh of terrain, facades, roofs, roads and vegetation in near-real time. Three groups each solved part of it. Commercial tools such as DJI FlightHub 2, Skydio X10 Onboard Modeling and SkyeBrowse produce fast previews or relative-accuracy models, but they publish no independently validated absolute accuracy for video input. Defence systems such as Vantor Raptor and Edgybees reach 1–3 m absolute accuracy by registering live video to prebuilt 3D or satellite reference data, not by building new 3D. Research from 2025–2026 now supplies the missing core: GPS-anchored feed-forward reconstruction (GeoFF3D, about 2,000 images in about 5 minutes, stated to be stable on near-collinear single-line trajectories), fast global SfM with GPS priors (COLMAP 4.x with GLOMAP merged in), and evidence that monocular SLAM holds about 0.5% horizontal drift on short nadir drone flights but fails badly in altitude. The accuracy literature points to one controlling fact: absolute error comes almost entirely from georeferencing. Consumer GNSS leaves a near-rigid offset of about 2 m horizontal and 4 m vertical, while RTK/PPK without GCPs reaches 2–5 cm. So a team that measures and removes that offset, and handles the doming that single strips cause, can reach the reported ≤1 m target. The realistic hackathon design has three layers. The geometry layer is keyframed, blur-filtered frames solved by GPS-prior SfM or GPS-anchored feed-forward reconstruction, then classical MVS for the metric mesh. A second layer reduces the absolute offset by registering to a public orthophoto or DSM, or by fusing RTK/CORS data, and reports CE90/LE90 honestly. A Gaussian-splat layer on top is for photoreal viewing only. At least ten public competitor repos already use the COLMAP + OpenMVS + GPS-alignment baseline, so a team stands out on absolute-accuracy engineering, degeneracy handling and evaluation discipline, not on having a pipeline.

No product ships single-pass video to metric 3D today

The commercial market splits cleanly into fast previews and slow, accurate pipelines, and none of the accurate ones is built for video.

DJI, the de facto hardware standard, has three rapid 3D paths, and all of them take stills or thumbnails, not video. DJI calls Terra's in-mission real-time 3D mapping "of low accuracy" and tells users to reconstruct again from the original images after landing (DJI Terra FAQ). It also warns that models "may be of poor quality or be unavailable in environments without RTK signals" (DJI Terra FAQ). FlightHub 2's February 2026 rapid modelling turns **up to 500 streamed thumbnails into a 3DGS or photogrammetry model in about 5 minutes**, with no published accuracy figure (DJI Enterprise blog). The Matrice 4E's Smart 3D Capture builds a rough model on the controller, and its purpose is to plan a close-proximity route, not to deliver a product (DJI Enterprise). DJI's accuracy benchmark, **3D models "within 4 cm" with no GCPs**, applies only to Phantom 4 RTK stills at 100 m processed in full after the flight. DJI's rule of thumb is accuracy of about 1–2× GSD (DJI Terra FAQ).

Skydio X10 Onboard Modeling builds a textured glTF mesh or an orthorectified map on the controller from 1920×1440 HD thumbnails. It processes about **30 images per minute** and finishes "within minutes after the flight" (Skydio support; Skydio blog). Skydio positions it as a coverage check, publishes no accuracy specification, and exports full-resolution data to Pix4D, Bentley and others for the real deliverable (Skydio X10). **Pix4D**'s rapid product, Pix4Dreact, makes 2D orthomosaics only, and from stills (Pix4D). Only the legacy PIX4Dmapper imports video, and it loses EXIF and camera auto-detection when it does (Pix4D support).

SkyeBrowse is the closest commercial analogue to PS 26158. It turns a "single, continuous video file" from a roughly two-minute orbit into a 3D model in "a few minutes" (The Drone Girl, 18 Sep 2026). It claims **0.5 in accuracy on Commercial/Pro tiers and 0.1 in with "16K Ultra"** processing, "without ground control points" (The Drone Girl; SkyeBrowse). These are vendor claims. No independent study was found, and "accuracy" here almost certainly means relative measurement accuracy on an orbit, not absolute georeferencing. An orbit is also the opposite of PS 26158's single linear pass: it gives converging views that avoid the single-strip degeneracy discussed below. **Agisoft Metashape** and **Bentley iTwin Capture** accept video, but they simply extract frames and feed a standard SfM pipeline. Metashape reportedly reads a matching DJI .SRT file for geotags (Agisoft forum; Wingtra on iTwin Capture).

Defence and intelligence work solves a different problem, and the difference is instructive. **Vantor (formerly Maxar) Raptor** positions drones and georegisters their full-motion video against an existing encrypted 3D terrain base covering more than 90 million km^2. It reaches **under 3 m target-coordinate accuracy (Sync/Ace) and under 10 m RMSE vision-based positioning (Guide)** without GNSS, but it does not generate 3D from the video (Vantor Raptor; Via Satellite). **Edgybees** georegisters aerial video to satellite imagery "in seconds" to within **1–2 m** (BAE GXP). The US Army's **One World Terrain / RAPTRS** (USC ICT) is the closest defence analogue to 3D generation. Soldiers fly an sUAS over about 1 km^2 and see it in 3D within **about 3 hours**, down from weeks, but this uses dedicated photogrammetric collection (army.mil). IARPA's **WRIVA** program (2023, 42 months) targets photorealistic 3D site models from very sparse ground, UAV and satellite imagery, with camera geolocation within 5 m (IARPA WRIVA).

The lesson from defence: **prior geospatial data plus live registration** gives metre-level absolute accuracy with no GCPs. A single-pass system can borrow this directly to fix its absolute offset.

Who	Input	Speed	Published accuracy	Gap vs PS 26158
DJI Terra real-time / FlightHub 2	Stills / ≤500 thumbnails	~5 min (FlightHub)	"Low accuracy" preview; none for FlightHub	Not video; accurate mode needs RTK + full re-processing
Skydio X10 Onboard	HD thumbnails	~30 img/min	None published	Preview only
SkyeBrowse	Single video (orbit)	Minutes	0.1–0.5 in (vendor, relative)	Orbit not linear pass; unvalidated
Metashape / iTwin / PIX4Dmapper / ODM	Video → frames + SRT	Hours	Depends on GCP/RTK	Batch, not near-real-time
Vantor Raptor / Edgybees	Live FMV + prior 3D/satellite	Seconds	<3 m / 1–2 m absolute	No new 3D generated
Army OWT/RAPTRS	Dedicated sUAS collect	~3 h per km^2	Not public	Not single-pass, not real-time

Georeferencing, not reconstruction, sets the accuracy ceiling

The most useful accuracy finding for this problem statement is that **consumer-GNSS error is almost entirely a rigid shift**. Zhang et al. flew at 45 m with 16 checkpoints and single-point GPS only, and measured RMSE of **2.29 m horizontal and 3.93 m vertical**. The vertical mean error was 3.93 m with a standard deviation of only 0.18 m, so the model's internal shape was near-perfect and it was simply displaced. A GoPro showed the same pattern: XY 1.53 m and Z 3.89 m, with an XY standard deviation of just 0.065 m (Zhang et al.

2019). With **PPK and zero GCPs**, the same study reached **XY 2.4 cm and Z 2.2 cm**, statistically equal to an 8-GCP solution. The authors still recommend one GCP to catch PPK biases and false fixes (Zhang et al. 2019). Other RTK/PPK studies without GCPs report horizontal 1–3 cm and vertical 4–6 cm on 71 checkpoints (Semantic Scholar PDF). The DJI/DroneDeploy whitepaper puts measured distances within **0.1–0.3%** of ground truth (DJI P4 RTK whitepaper). The consequence for PS 26158 is direct. Relative metrics such as lengths, areas and volumes are achievable from consumer GPS today. Absolute ≤1 m needs one extra anchor: a known point, a reference orthophoto or DSM, a SBAS/CORS correction, or RTK/PPK data (listed as an optional input in the PS).

Video frames can match stills if they are sampled densely enough. A 2021 comparison found building-boundary RMSE of **1.1–2.7 cm for video sampled at one frame per 2.5 s**, versus 1.3–2.8 cm for a photo flight. Accuracy degraded as the interval grew to 4 s and 5.5 s (Remote Sens. 2021, 13(14), 2745; abstract-level numbers). The costs of video are real. 4K frames are only about 8 MP against 20 MP stills, and compression, motion blur and rolling shutter all add error (PhotoModeler). Rolling shutter matters most for consumer drones. Pix4D's Vautherin et al. measured **74 ms readout on a Phantom 2 Vision+ and 30 ms on an Inspire 1**. At 8 m/s and 70 m altitude that displaces image content by about 10 px and 4 px respectively, and modelling it removed systematic reprojection error and markedly improved checkpoint accuracy (Vautherin et al. 2016). Time synchronisation is the other quiet killer. For cm-level direct georeferencing at 10–30 m/s, exposure-to-GNSS sync must be **better than 1 ms** (ResearchGate: time sync on MAVs). DJI's SRT sidecar logs latitude, longitude, altitude and focal length per frame, but in local time with no timezone (ScienceDirect 2025), and no study was found that measures its actual latency against exposure.

A single pass is the worst geometry photogrammetry knows. James & Robson showed that self-calibrating bundle adjustment on near-parallel nadir images cannot recover radial distortion. The result is a **dome or bowl deformation of metre magnitude in simulation and about 0.2 m in a real test** (James & Robson 2014). The fixes they identified are counterintuitive. A second set of flight lines at a different azimuth did *not* help much. Varying camera pointing or altitude did, and oblique images at 20–30° **cut DEM deformation by one to two orders of magnitude**. Fixing K1 to a doming-free value gave a tenfold reduction (James & Robson 2014). A newer recommendation is oblique images at 20–45° making up at least 10% of the block (Drones 2025, 9(6), 387). For corridors, a pre-calibrated camera with GNSS-assisted orientation and **no GCPs reached 5.9/3.3/7.0 cm (X/Y/Z)** over a 1,200 × 180 m strip (Sensors 2018). A 2026 paper adds "sub-scene-based GNSS-constrained SfM" to keep degenerate stretches of a long strip from corrupting camera parameters (MDPI Remote Sensing 18(14):2321). There is also a geometric blind spot, which is a derived inference rather than a sourced result. GPS positions along a straight line are collinear, so aligning to them cannot fix roll about the flight axis. That angle must come from gimbal attitude, IMU gravity, a curved path or altitude variation.

Evaluators will probably measure accuracy the way ASPRS Edition 2 (2023) specifies: **RMSE only, at least 30 independent checkpoints, each at least twice as accurate as the product** (LIDAR Magazine). A defence audience will expect CE90 and LE90, where **LE90 = 1.6449 × RMSE_Z** and CE90 is about 2.146 × per-axis RMSE (Maxar accuracy whitepaper). Converting Zhang's numbers gives about **3.5 m CE90 and 6.5 m LE90 for uncorrected consumer GNSS**, which fails a ≤1 m target. This is a derived conversion. One 2026 UAV study shows the same gap for Gaussian splatting built straight from video frames with GNSS metadata. Internal consistency was 4.8 mm ICP RMS, but absolute accuracy was **5.26 m CE90** (T&F 2026).

GPS-anchored feed-forward models beat pure SLAM on single passes

The best single-pass-specific evidence is a 2026 benchmark by Spagnolo, Dobrevski and Skočaj. It ran five monocular SLAM systems on high-altitude nadir drone footage, including five consumer DJI flights of 1–2 km at 110–120 m. On those short flights, **MASt3R-SLAM drifted 0.53% and DROID-SLAM 0.58% of path length horizontally**. ORB-SLAM3 drifted 2.40%, DPVO 5.75% and VGGT-SLAM 2.0 6.34% (arXiv 2608.18632). Longer flights broke everything. **Vertical RMSE exceeded 100 m in 29 of 34 long runs**, and on the longest helicopter sequences "no method yields a usable trajectory" (arXiv 2608.18632). The failures differ by method. ORB-SLAM3 loses tracking, DROID-SLAM drifts gradually, DPVO wanders out of the survey area, and MASt3R-SLAM fails on homogeneous terrain. Compute matters for a hackathon too. DROID-SLAM, MASt3R-SLAM and VGGT-SLAM 2.0 each needed about **30 GB of GPU memory**, DPVO under 3 GB, and only ORB-SLAM3 runs on CPU. The authors prescribe inertial fusion, altimetry, absolute localisation against reference imagery, and aerial-adapted priors (arXiv 2608.18632). A single pass never revisits a location, so loop closure has nothing to correct. Drift control has to come from GPS, barometer and reference imagery. The classical way to fuse these is GNSS-visual-inertial factor graphs such as GVINS, which tightly couples raw pseudoranges with visual and inertial factors (arXiv 2103.07899). However, consumer drone video files usually lack the synchronised raw IMU stream these systems need (VINS-Mono).

Feed-forward 3D foundation models changed the picture in 2025–2026, but only once they were adapted to aerial data. The first systematic evaluation of DUSt3R, MASt3R and VGGT on photogrammetric aerial blocks tested fewer than 10 images at 518 px. It found **up to 50% more dense completeness than COLMAP**, with VGGT the most efficient. Pose reliability dropped as resolution and image count grew, and the authors concluded the models "cannot fully replace traditional SfM and MVS" (Wu et al., arXiv 2507.14798). **UAVFF3D** (May 2026) named three aerial failure modes: appearance shift, oblique views, and **ambiguity between field of view and flying height**. Adapting models on 170k real and 370k synthetic UAV images cut pose ATE by up to 76% and Chamfer distance by up to 41% (arXiv 2605.17942).

The same group's **GeoFF3D** (August 2026) comes closest to PS 26158 of any published method. It feeds georeferenced camera translations (GPS) into the network. It predicts poses and dense point maps **directly in a gravity-aligned, Z-up metric frame**, processing overlapping chunks and merging them hierarchically. It reconstructs **about 2,000 images in about 5 minutes** and is described as stable on near-collinear trajectories. On long UAVScenes sequences it scored **F@5 = 0.848, against 0.687 for Pi3X and 0.451 for a SLAM/streaming baseline** (arXiv 2608.28288; GitHub).

MapAnything (Meta/CMU) also accepts optional intrinsics and poses and regresses metric geometry, so drone telemetry can be fed in (arXiv 2509.13414). **VGGT-Long** shows how to chunk calibration-free reconstruction to kilometre scale with Sim(3) alignment, though it was evaluated only on driving data (arXiv 2507.16443). Licences decide what NTRO can deploy. DUSt3R and MASt3R are non-commercial. The original VGGT checkpoint is non-commercial, and **VGGT-1B-Commercial explicitly excludes military use** (VGGT GitHub). GeoFF3D, MapAnything, π3 and DA3 licences were not verified and must be checked before a defence pitch.

Monocular metric depth is not the scale source. The ECCV 2026 **AerialMetric** benchmark (52k real and 16k synthetic UAV pairs) found "significant domain gaps" for street- and indoor-trained models, which need aerial fine-tuning (arXiv 2606.29716). At 100 m or more, depth varies little relative to range, so scale is better taken from GPS baselines and altitude. Depth models are better used as densification priors.

Classical SfM remains the accuracy reference, and it has become much faster. **GLOMAP** global SfM is "orders of magnitude faster" than incremental COLMAP at comparable accuracy and handles sequential video (arXiv 2407.20219). It is now merged into COLMAP as the "global" mapper (GLOMAP README). The COLMAP 4.x releases add **GPS and gravity pose priors, ALIKED/LightGlue features, and GPU bundle adjustment billed as 1–2 orders of magnitude faster** (COLMAP releases; the fetched release dates were internally inconsistent, so verify versions before quoting them).

Gaussian splats look better but measure worse

Gaussian splatting has overtaken NeRF for large aerial view synthesis. On Mill-19 and UrbanScene3D, DOGS and VastGaussian train in **about 2.5–4.5 hours versus 18–23 hours for vanilla 3DGS**, with PSNR differences between modern methods under 1 dB (DOGS, arXiv 2405.13943). Rendering quality and geometric accuracy are separate things, though. Surface-oriented variants such as PGSR (0.52 mm DTU Chamfer in 0.5 h) and CityGaussianV2 (F1 0.556 on MatrixCity-Aerial) extract meshes by TSDF. The city-scale benchmarks use **1 m voxels**, far coarser than photogrammetric meshes (PGSR; CityGaussianV2). An ISPRS 2026 metric assessment found 3DGS geometry "generally slightly lower than traditional photogrammetry," especially with small datasets or low-resolution input, which describes single-pass video (ISPRS Archives 2026). The defensible design is MVS for measurement and GS for viewing. A live UAV-video GS pipeline (RTMP stream, sensor fusion, continuous updates) reached quality within 4–7% of offline references but reported no geometric comparison (arXiv 2602.20342). SLAM-plus-GS systems mostly run at 0.3–3.7 fps and are tested indoors (SplatMAP; GaussianFlow SLAM).

Each named challenge in PS 26158 has a matching method.

Dynamic objects. DroneSplat (CVPR 2025) is the only aerial-specific method. It uses SAM v2 adaptive local-global masking, including video tracking to catch temporarily parked vehicles, and gains **+0.9 to +2.4 dB PSNR** over 3DGS and WildGaussians on 24 drone scenes. SAM v2 becomes a compute bottleneck on large scenes, and DroneSplat does not handle lighting changes (arXiv 2503.16964; GitHub). SpotLessSplats and WildGaussians do the same job with semantic features and robust losses (arXiv 2406.20055). In a single pass a moving car appears in few frames, so robust residual masks work. Class masks from a detector applied before SfM are a cheap, deterministic addition. Their gain on aerial benchmarks has not been quantified.

Motion blur and rolling shutter. BAD-Gaussians models blur as integration along the camera path during exposure (project). Spectacular AI's **"Gaussian Splatting on the Move"** (ECCV 2024) jointly compensates motion blur and rolling shutter using VIO velocity, and its code is open (GitHub). No verified aerial-specific numbers exist for either, so sharpness-based keyframe rejection is the cheap first defence.

Illumination. Per-image affine exposure compensation, as in PGSR, and decoupled appearance modelling, as in VastGaussian, absorb auto-exposure drift (PGSR; VastGaussian). A minutes-long pass sees little change in sun angle, so exposure drift is the real issue. Aerial shadow removal remains unsolved.

Occlusion. A single pass under-observes facades and roof overhangs. Diffusion completion (ReconFusion, CAT3D, ViewCrafter, Difix3D+) produces plausible but non-metric content (GSFix3D). Structured building priors are metric. **City3D** infers missing vertical walls from roof planes and was released with 20,000 buildings (arXiv 2201.10276), and City4CFD produces LoD2 models (GitHub). Any synthesised surface should be flagged in the deliverable.

Delivery. The web delivery standard now covers both layers. Khronos added Gaussian splats to glTF (KHR_gaussian_splatting). **CesiumJS 1.139 and Cesium for Unreal 2.23 shipped support in March 2026**, followed by hierarchical LOD for splat 3D Tiles in April 2026 (Khronos; radiancefields.com; Cesium blog). A measurable mesh and a photoreal splat layer can now share one viewer.

Open-source parts exist but are batch-grade or abandoned

OpenDroneMap is the practical baseline. Since v3.0.4 it ingests .mp4/.mov/.ts video with matching DJI .srt GPS directly, capped by default at **500 extracted frames (`--video-limit`)**, and outputs GeoTIFF orthophoto, DSM/DTM, LAZ, OBJ/PLY and **OGC 3D Tiles (`--3d-tiles`)** (ODM GitHub; ODM arguments). Recent releases add rolling-shutter profiles for DJI Mini 3, Mini 4 Pro and Mavic 3, and SRT parsing fixes (ODM releases). It is slow. A community benchmark measured **242 images at ultra quality in 124 min on CPU and 62 min with GPU**, and 473 images in 316 min with GPU, peaking near 40 GB of RAM (ODM community). That is far outside a 15-minute budget unless quality is cut and the dense stage replaced. It is also AGPL-3.0, which matters for a hosted service.

OpenREALM (TU Braunschweig) is the only open real-time UAV mapper found. It combines visual SLAM, GPU plane-sweep stereo, a live 2.5D elevation map and an incremental orthophoto, and it has run on a Jetson Xavier NX (GitHub; Springer PFG). It is self-described alpha ROS1 code whose last significant update was around 2020. Lighter CPU approaches reach **2.3 fps near-real-time mosaicking** (ISPRS Annals 2025; MDPI Appl. Sci. 2026). A live 2D preview is therefore easy. Live metric 3D is not.

Telemetry ingest is solved in pieces. There are many DJI SRT parsers, such as DJI_SRT_Parser and dji-telemetry, but SRT formats vary by model. For military FMV, **klvdata** decodes MISB ST 0601 metadata from STANAG 4609 streams (PyPI klvdata). Few student teams will support this, and it matters for an NTRO audience.

For evaluation, **MARS-LVIG** is the best public stand-in for a single-pass flight. It has 21 downward-looking sequences at 80–130 m and 3–12 m/s with a global-shutter camera, IMU, raw GNSS, **RTK ground-truth poses and a DJI L1 LiDAR map** (IJRR; download). UseGeo provides ground-truth depth and LiDAR point clouds (GitHub). UAVD4L provides an RTK/GCP-aligned textured mesh and DSM over about 2.5 km^2 for testing registration to a prior model (arXiv 2401.05971).

India's constraints favour an offline, CORS-aware tool

The PS itself frames the job. SIH26158 (NTRO, Software, Robotics & Drones) asks for a georeferenced, metric, textured model from one 1080p/4K pass plus GPS and flight metadata. IMU, barometer, intrinsics and RTK/PPK are optional inputs. The listed challenges are limited viewing angles, blur, lighting, dynamic objects, GPS inaccuracy, real-time load, occlusion, and metric accuracy without extensive GCPs (zaidsayyed.in SIH26158). Team repos cite targets of **under 15 min of processing for a 10-min video and ≤1 m accuracy**, with outputs in OBJ, PLY, LAS, GeoTIFF, glTF and FBX (Tesseract repo; Scout-3D). One mirror, however, says the listing has no explicit numbers, so **verify these targets on sih.gov.in** before designing to them.

The competition is already visible. At least ten public repos exist. The most mature, **Tesseract**, runs shot-cut filtering, optical-flow keyframing, MapAnything focal estimation, COLMAP's global mapper with fixed intrinsics, and OpenMVS, plus custom seam levelling. It adds SRT-based GPS alignment, gravity-fixed roll and

pitch, and an "honest output" manifest linking each accuracy claim to evidence. It reports **548 s for a synthetic 10-min flight and 0.344 m median error on synthetic data** (Tesseract). Others, such as noob-rishav, replace GCPs with GPS+IMU. The shared baseline is COLMAP/GLOMAP + OpenMVS + GPS alignment. Honest reporting alone is no longer novel. The open ground is real-flight absolute accuracy against independent checkpoints.

Indian rules shape the architecture. The **DST 2021 geospatial guidelines** require that data finer than **1 m horizontal or 3 m vertical** be owned by Indian entities and stored and processed in India (DST guidelines). A system that hits its ≤1 m target therefore produces regulated data, which argues for an offline or on-premise pipeline with no foreign cloud APIs. That suits NTRO, India's PMO/NSA-linked technical intelligence agency covering IMINT and GEOINT (Wikipedia NTRO). Absolute anchors are available domestically. **Survey of India CORS** offers RTK/NRTK at **±3 cm** from roughly 1,000+ stations, free for government and academia (SoI CORS; Geospatial World). **GAGAN SBAS** gives about **1.5 m horizontal and 2.5 m vertical** (Wikipedia GAGAN). NavIC alone gives about 5–20 m (Wikipedia IRNSS). The national accuracy benchmark is SVAMITVA, about **3.3 lakh villages flown at 1:500 scale with ±5 cm** using RTK/PPK drones and CORS (PIB; ETV Bharat). PS 26158's ≤1 m is a rapid-response tier, not cadastral. Under the Drone Rules 2021, demo flights must stay in **green zones (up to 120 m)**. Border and sensitive sites are red zones, so the tool must work on pre-recorded video from authorised operators (PIB Drone Rules).

A buildable architecture that wins on accuracy engineering

The evidence supports a pipeline that treats **relative geometry and absolute placement as separate problems** and reports each honestly. The table below is the recommended build. Each stage is chosen from the evidence above, and the time budget targets a 10-minute 4K clip on a single RTX-class workstation. The per-stage timings are estimates to measure, not sourced figures.

Stage	What to build	Why (evidence)
0. Ingest	ffmpeg decode; DJI SRT parser **and** MISB 0601 KLV via klvdata; per-frame GPS/alt/gimbal/focal; estimate the SRT-to-frame time offset by cross-correlating GPS velocity with SfM velocity	SRT has no timezone and unmeasured latency; ms-level sync matters; KLV is the defence FMV format
1. Keyframes	Parallax/optical-flow keyframing at about 1–2 fps (about 600–1,200 frames for 10 min), variance-of-Laplacian blur rejection per window, shot-cut/overlay masking	1 frame per 2.5 s matched stills; denser adds compute, not accuracy
2. Masking	Detector/segmenter masks for vehicles and people before matching; DroneSplat-style residual masks later	Removes dynamic features from SfM and texture
3. Poses (primary)	COLMAP 4.x global mapper + ALIKED/LightGlue + **GPS pose priors** + gravity/gimbal priors; **rolling-shutter model**; intrinsics fixed from a pre-calibration or tightly constrained (never free K1 on a nadir strip); sub-scene segmentation on long strips	Doming, roll-about-track and strip degeneracy fixes
3b. Poses (fast/fallback)	GeoFF3D (GPS-anchored chunks) or MapAnything with pose/intrinsic inputs; DPVO as a low-memory local-pose fallback	2,000 imgs in ~5 min; stable on collinear paths; DPVO <3 GB. **Check licences**

Stage	What to build	Why (evidence)
4. Absolute anchoring	Robust 7-DoF alignment to GPS, then **register to a reference orthophoto/DSM** (the Raptor/Edgybees pattern), with optional ingest of PPK/RTK or CORS/GAGAN-corrected positions and a single optional GCP	Consumer GNSS error is a rigid ~2–4 m shift; one anchor removes most of it
5. Dense + mesh	OpenMVS densify → mesh → texture with seam levelling at medium resolution; DSM/orthophoto rasterisation	MVS geometry beats GS geometry
6. Completion	Plane-based LoD2 roof and wall fitting (City3D-style) for occluded facades, flagged as inferred	Metric, CityGML-friendly
7. Photoreal layer	gsplat/PGSR-style GS with exposure compensation, trained in the background	Viewing only
8. Delivery	3D Tiles mesh + KHR_gaussian_splatting layer in CesiumJS; LAS/LAZ, GeoTIFF, OBJ/glTF export; per-vertex **confidence/coverage heatmap**	One viewer, measurable plus photoreal
9. QA report	CE90/LE90, RMSE_H/RMSE_Z, bias vs noise split, relative length error; tested on MARS-LVIG (RTK + L1 LiDAR truth) and one real green-zone flight with ≥30 RTK checkpoints	ASPRS Ed. 2 practice; evaluator language

Several design choices here are aimed at differentiation, because the baseline is commoditised. First, **attack the absolute offset explicitly.** Show the Zhang-style decomposition, in which the error is mostly bias. Then show that orthophoto/DSM registration or a CORS/GAGAN correction collapses it. Report before-and-after CE90 on real checkpoints. No public competitor repo demonstrates real-flight absolute accuracy. Second, **own the single-strip degeneracy.** Estimate time offset and rolling shutter, lock intrinsics, recover roll from gimbal and gravity, and publish a doming test, such as a DSM residual against the MARS-LVIG LiDAR along and across track. Add a short capture guideline (forward-oblique gimbal at 20–30°, gentle altitude variation), because the literature shows this reduces deformation by one to two orders of magnitude at zero compute cost. Third, **speak NTRO's language.** Support KLV/STANAG 4609 ingest, run fully offline and air-gapped in line with the DST data-localisation rule, use only licence-clean models (not VGGT-Commercial, which bars military use), and quote CE90/LE90. Fourth, **show uncertainty.** A coverage and confidence map that marks under-observed facades and LoD2-inferred surfaces answers the occlusion challenge honestly instead of hiding it behind a pretty splat. Finally, show a **live 2D preview** during processing, since incremental CPU mosaicking at about 2 fps is proven. Deliver the metric 3D product minutes later. Do not claim real-time 3D that the literature does not support.

Several uncertainties remain for the team to resolve. The official PS text and its accuracy and time targets need confirmation on sih.gov.in. GeoFF3D's licence, hardware for its 5-minute figure, and behaviour on low-altitude oblique video are unverified. No peer-reviewed benchmark exists for forward-looking, oblique single-pass drone video with ground truth. DJI SRT timing latency has never been measured publicly. Relative accuracy tied to a GPS-only scale should improve with trajectory length and averaging. A rough derivation suggests about 0.1% scale error over a 300 m path, consistent with DJI's measured 0.1–0.3% distance agreement, but correlated GNSS errors make this optimistic.

Conclusion

The gap PS 26158 targets is real, but it is narrower than it looks. Reconstruction from a single pass is essentially solved at the relative level. Learned and classical methods hold sub-percent drift and centimetre

internal consistency over the 1–2 km scales a drone pass covers. What remains unsolved is the join between that geometry and the Earth: an unmeasured SRT time offset, rolling shutter, a doming-prone strip geometry, and a consumer-GNSS bias of several metres. Commercial vendors sidestep this with RTK hardware and re-processing. Defence vendors sidestep it with prior 3D data. Neither has published a validated answer for plain video. The team that treats georeferencing as the core engineering problem, rather than a final alignment step, is addressing the part of the problem that is actually open.

This also changes what the prototype should prove. A beautiful mesh is table stakes; ten repos already produce one. The persuasive demo is a before-and-after table on independent checkpoints: raw GPS placement at several metres CE90, then anchored placement at sub-metre, with doming residuals and confidence maps beside it, all running offline within the time budget. That kind of evidence travels from a hackathon stage into an NTRO evaluation. Its shape follows the one path that has actually moved from SIH to government adoption, ISRO's acceptance of a video-geotagging tool (PIB SIH 2024): a practical, measurable, deployable geospatial utility.